WORKED SOLUTION

STEP III 2000, Question 5

Pure • Linear Transformations

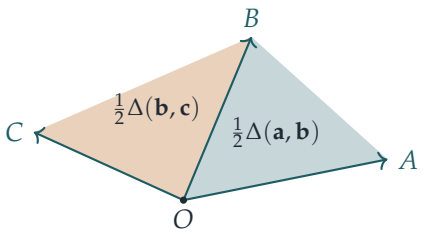
STEP III 2000, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q5

The key idea

$\Delta(\mathbf{a}, \mathbf{b})$ is the determinant of the matrix whose columns are $\mathbf{a}$ and $\mathbf{b}$, so it is twice the signed area of the triangle OAB, and it is non-zero exactly when $\mathbf{a}$ and $\mathbf{b}$ are not parallel. A linear map multiplies every such area by the same factor, its determinant. That one fact gives necessity at once. For sufficiency, the identity in (ii) writes $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ using only Δ's, so once $\mathbf{M}$ is pinned down by A and B, it has no choice about where C goes.



Solution

Throughout, write $[\mathbf{u}\ \mathbf{v}]$ for the 2×2 matrix with columns $\mathbf{u}$ and $\mathbf{v}$. Then

$$\Delta(\mathbf{u}, \mathbf{v}) = u_1 v_2 - u_2 v_1 = \det[\mathbf{u}\ \mathbf{v}].$$

For non-zero $\mathbf{u}$, $\mathbf{v}$, we have $\Delta(\mathbf{u}, \mathbf{v}) = 0$ exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel: the determinant vanishes precisely when one column is a multiple of the other. So all six of $\Delta(\mathbf{a}, \mathbf{b})$, $\Delta(\mathbf{b}, \mathbf{c})$, $\Delta(\mathbf{c}, \mathbf{a})$, $\Delta(\mathbf{p}, \mathbf{q})$, $\Delta(\mathbf{q}, \mathbf{r})$, $\Delta(\mathbf{r}, \mathbf{p})$ are non-zero.

Two more facts get used repeatedly. Since $\mathbf{M}[\mathbf{u}\ \mathbf{v}] = [\mathbf{M}\mathbf{u}\ \ \mathbf{M}\mathbf{v}]$ (column by column) and determinants multiply,

$$\Delta(\mathbf{M}\mathbf{u}, \mathbf{M}\mathbf{v}) = \det \mathbf{M}\ \Delta(\mathbf{u}, \mathbf{v}). \qquad (*)$$

And $\Delta(\mathbf{v}, \mathbf{u}) = -\Delta(\mathbf{u}, \mathbf{v})$.

Part (i)

We need $\mathbf{Ma} = \mathbf{p}$ and $\mathbf{Mb} = \mathbf{q}$, which together say

$$\mathbf{M}\,[\mathbf{a}\ \mathbf{b}] = [\mathbf{p}\ \mathbf{q}].$$

Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel, $\det[\mathbf{a}\ \mathbf{b}] = \Delta(\mathbf{a},\mathbf{b}) \neq 0$, so $[\mathbf{a}\ \mathbf{b}]$ is invertible, and

$$\mathbf{M} = [\mathbf{p}\ \mathbf{q}]\,[\mathbf{a}\ \mathbf{b}]^{-1} = \frac{1}{\Delta(\mathbf{a},\mathbf{b})}\begin{pmatrix} p_1 & q_1 \\ p_2 & q_2 \end{pmatrix}\begin{pmatrix} b_2 & -b_1 \\ -a_2 & a_1 \end{pmatrix}$$

does the job. In fact this $\mathbf{M}$ is the *only* such matrix, because we can multiply on the right by the inverse. Taking determinants, $\det\mathbf{M} = \Delta(\mathbf{p},\mathbf{q})/\Delta(\mathbf{a},\mathbf{b})$, which is non-zero.

Part (ii): the identity

Take the first component of the left-hand side:

$$\begin{aligned}&(a_1b_2 - a_2b_1)c_1 + (c_1a_2 - c_2a_1)b_1 + (b_1c_2 - b_2c_1)a_1\\ &\quad = a_1b_2c_1 - a_2b_1c_1 + a_2b_1c_1 - a_1b_1c_2 + a_1b_1c_2 - a_1b_2c_1 = 0,\end{aligned}$$

since the six terms cancel in pairs. The second component is the same calculation with every subscript on $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ outside the Δ's changed from 1 to 2:

$$\begin{aligned}&(a_1b_2 - a_2b_1)c_2 + (c_1a_2 - c_2a_1)b_2 + (b_1c_2 - b_2c_1)a_2\\ &\quad = a_1b_2c_2 - a_2b_1c_2 + a_2b_2c_1 - a_1b_2c_2 + a_2b_1c_2 - a_2b_2c_1 = 0.\end{aligned}$$

So

$$\Delta(\mathbf{a},\mathbf{b})\,\mathbf{c} + \Delta(\mathbf{c},\mathbf{a})\,\mathbf{b} + \Delta(\mathbf{b},\mathbf{c})\,\mathbf{a} = \mathbf{0}, \tag{1}$$

as required. The identity holds for *any* three vectors, so it holds for $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ too:

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{r} + \Delta(\mathbf{r},\mathbf{p})\,\mathbf{q} + \Delta(\mathbf{q},\mathbf{r})\,\mathbf{p} = \mathbf{0}. \tag{2}$$

? **Ask yourself** What exactly does the ratio condition say, as an equation?

All the Δ's are non-zero, so the condition $\Delta(\mathbf{a},\mathbf{b}) : \Delta(\mathbf{b},\mathbf{c}) : \Delta(\mathbf{c},\mathbf{a}) = \Delta(\mathbf{p},\mathbf{q}) : \Delta(\mathbf{q},\mathbf{r}) : \Delta(\mathbf{r},\mathbf{p})$ means there is a single number $k \neq 0$ with

$$\Delta(\mathbf{p},\mathbf{q}) = k\,\Delta(\mathbf{a},\mathbf{b}), \qquad \Delta(\mathbf{q},\mathbf{r}) = k\,\Delta(\mathbf{b},\mathbf{c}), \qquad \Delta(\mathbf{r},\mathbf{p}) = k\,\Delta(\mathbf{c},\mathbf{a}). \tag{3}$$

In words: the three signed triangle areas OPQ, OQR, ORP are the areas OAB, OBC, OCA all scaled by the same factor.

Part (ii): necessity

Suppose $\mathbf{Ma} = \mathbf{p}$, $\mathbf{Mb} = \mathbf{q}$ and $\mathbf{Mc} = \mathbf{r}$. By $(*)$,

$$\Delta(\mathbf{p},\mathbf{q}) = \det\mathbf{M}\,\Delta(\mathbf{a},\mathbf{b}), \quad \Delta(\mathbf{q},\mathbf{r}) = \det\mathbf{M}\,\Delta(\mathbf{b},\mathbf{c}), \quad \Delta(\mathbf{r},\mathbf{p}) = \det\mathbf{M}\,\Delta(\mathbf{c},\mathbf{a}).$$

This is (3) with $k = \det\mathbf{M}$, and $k \neq 0$ because $\Delta(\mathbf{p},\mathbf{q}) \neq 0$. So the ratios are equal.

Part (ii): sufficiency

Now suppose (3) holds for some $k \neq 0$. By part (i) there is a matrix $\mathbf{M}$ with $\mathbf{Ma} = \mathbf{p}$ and $\mathbf{Mb} = \mathbf{q}$; we must show that this same $\mathbf{M}$ also sends $\mathbf{c}$ to $\mathbf{r}$.

Apply $\mathbf{M}$ to (1). Matrix multiplication is linear and the Δ's are just numbers, so

$$\Delta(\mathbf{a},\mathbf{b})\,\mathbf{Mc} = -\Delta(\mathbf{c},\mathbf{a})\,\mathbf{q} - \Delta(\mathbf{b},\mathbf{c})\,\mathbf{p}.$$

Multiply by k and use (3) on each term:

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{Mc} = -\Delta(\mathbf{r},\mathbf{p})\,\mathbf{q} - \Delta(\mathbf{q},\mathbf{r})\,\mathbf{p}.$$

By (2), the right-hand side equals $\Delta(\mathbf{p},\mathbf{q})\,\mathbf{r}$. So

$$\Delta(\mathbf{p},\mathbf{q})\,\mathbf{Mc} = \Delta(\mathbf{p},\mathbf{q})\,\mathbf{r},$$

and since $\Delta(\mathbf{p},\mathbf{q}) \neq 0$ (as $\mathbf{p}$, $\mathbf{q}$ are not parallel), $\mathbf{Mc} = \mathbf{r}$. Hence P, Q, R are the images of A, B, C under $\mathbf{M}$.

Necessity and sufficiency together prove that the ratio condition is necessary and sufficient.

Checks

- Try $\mathbf{a} = \binom{1}{0}$, $\mathbf{b} = \binom{0}{1}$, $\mathbf{c} = \binom{1}{1}$. Then $\Delta(\mathbf{a},\mathbf{b}) = 1$, $\Delta(\mathbf{b},\mathbf{c}) = -1$, $\Delta(\mathbf{c},\mathbf{a}) = -1$, and (1) reads $\binom{1}{1} - \binom{0}{1} - \binom{1}{0} = \mathbf{0}$. With $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ we get $\mathbf{p} = \binom{2}{0}$, $\mathbf{q} = \binom{1}{3}$, $\mathbf{r} = \binom{3}{3}$, and the three Δ's are 6, -6, -6: the ratio $1 : -1 : -1$ again, with $k = 6 = \det \mathbf{M}$.
- Numerically, we tested the identity on 20,000 random integer triples, using exact fractions. For about 17,700 triples with no two vectors parallel, we built $\mathbf{M}$ from part (i) and confirmed necessity, and that $\mathbf{Mc} = \mathbf{r}$ exactly when (3) holds.

Common slips

- Writing the ratio condition as one equation of fractions without saying the Δ's are non-zero. The hypotheses that no two vectors are parallel are there precisely to make every division legal.
- Proving only necessity. The real content is sufficiency: a matrix is fixed by its action on two independent vectors, so the condition must force the third image.
- Sign errors from the order inside Δ: the cyclic order $\mathbf{b},\mathbf{c}$ then $\mathbf{c},\mathbf{a}$ matters, since $\Delta(\mathbf{a},\mathbf{c}) = -\Delta(\mathbf{c},\mathbf{a})$.

Beyond STEP

Identity (1) is Cramer's rule in disguise: it gives the coordinates of $\mathbf{c}$ in the basis $\mathbf{a}$, $\mathbf{b}$ as ratios of determinants. The fact $(*)$, that a linear map scales every signed area by $\det \mathbf{M}$,

is the geometric definition of the determinant, and it is behind the Jacobian factor in a change of variables for double integrals. Strang's *Introduction to Linear Algebra* develops both ideas.

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